# Report for the Course on Cyber-Physical Systems and IoT Security

## Second mid-term project

Professional
Security
Corporation


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**Reference paper**

## An Anomaly-Based IDS for Detecting Attacks in RPL-Based Internet of Things

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# 1 Introduction

The Internet of Things (IoT) has evolved into a widespread paradigm connecting large numbers of resource-constrained devices and enabling the development of intelligent cyber-physical systems. Many IoT deployments rely on Low-Power and Lossy Networks (LLNs), where nodes operate under strict limitations in terms of processing power, memory, energy, and communication bandwidth. Technologies such as IPv6 over Low-Power Wireless Personal Area Networks (6LoWPAN) enable IPv6 communication in these constrained environments, while the Routing Protocol for Low-Power and Lossy Networks (RPL), standardized by the Internet Engineering Task Force (IETF), provides distributed routing and topology formation.

The characteristics of LLNs and the distributed nature of RPL introduce significant security challenges. In particular, compromised internal nodes can legitimately participate in the routing process while manipulating RPL control information, making conventional security mechanisms insufficient against internal routing attacks [1]. Among these attacks, Sinkhole and Blackhole attacks are particularly relevant. A Sinkhole node can advertise an artificially attractive routing metric to attract network traffic through itself, while a Blackhole node can exploit its position in the routing topology to selectively or completely discard forwarded packets.

This work investigates the behavior of a RPL network under Sinkhole and Blackhole attacks and evaluates their impact on network performance. In addition, a lightweight log-based Intrusion Detection System (IDS) is developed to identify routing anomalies by monitoring RPL control messages and detecting inconsistencies in node and parent ranks.

# 2 Background

This section introduces the concepts required to understand the experimental work.

## 2.1 The RPL Routing Protocol

The RPL is a distance-vector routing protocol designed for resource-constrained LLNs. RPL organizes nodes into a Destination-Oriented Directed Acyclic Graph (DODAG), whose topology is rooted at a node providing connectivity towards an external network.

### 2.1.1 DODAG Construction and Topology Management

Each node in a DODAG is assigned a *Rank*, which represents its position relative to the root. The root has the lowest Rank, while nodes farther from the root generally have higher Rank values. Rank is used to maintain the directed acyclic structure of the DODAG and to support loop avoidance.

The selection of a preferred parent and the computation of the resulting Rank are determined by an *Objective Function (OF)*, which evaluates the routing cost associated with neighboring nodes. Two OFs commonly associated with RPL are:

- **Objective Function Zero (OF0):** primarily considers the number of hops towards the root when selecting a parent;
- **Minimum Rank with Hysteresis Objective Function (MRHOF):** considers link-quality metrics such as the Expected Transmission Count (ETX) to compute the path cost and applies hysteresis to limit unnecessary parent changes.

In the RPL implementation considered in this work, MRHOF is used for parent selection. The advertised Rank of a neighbor therefore represents an important component of the routing cost considered when evaluating potential parents.

### 2.1.2 RPL Control Messages

RPL uses Internet Control Message Protocol for IPv6 (ICMPv6) control messages to construct and maintain the DODAG:

- **DODAG Information Object (DIO):** advertises DODAG information, including the current Rank, version, and other configuration parameters. Nodes use received DIOs to discover potential parents;
- **DODAG Information Solicitation (DIS):** requests DIO messages from neighboring nodes, for example when joining or recovering from a disconnected state;

- **Destination Advertisement Object (DAO):** advertises routing information towards the root, allowing downward routes to be maintained;
- **Destination Advertisement Object Acknowledgment (DAO-ACK):** optionally acknowledges the reception of a DAO.

Among these messages, the DIO is particularly relevant to the attacks considered in this work, since it contains the Rank advertised by a node and is therefore involved in parent selection.

## 2.2 Routing Attacks in RPL Networks

The distributed nature of RPL allows a compromised internal node to participate in the routing protocol while providing malicious routing information. Two routing attacks are considered in this project: Blackhole and Sinkhole.

### 2.2.1 Blackhole Attack

A Blackhole attack consists of a compromised routing node that discards packets that it is expected to forward instead of delivering them towards their destination. The malicious node may continue to participate normally in RPL topology formation, while disrupting data traffic that is routed through it.

The attack primarily affects packet forwarding and can therefore result in packet loss and reduced network delivery performance.

### 2.2.2 Sinkhole Attack

A Sinkhole attack attempts to attract network traffic by making a compromised node appear to provide a highly favorable route towards the DODAG root. In RPL, this can be achieved by manipulating the Rank advertised in DIO messages.

The attacker advertises a Rank lower than its actual Rank, causing neighboring nodes to potentially consider it a more attractive parent. Once selected as a preferred parent, the malicious node can attract traffic originating from other nodes in its neighborhood.

The effectiveness of the attack depends on the routing OF, link metrics, and parent-selection mechanisms. In particular, the hysteresis mechanism of MRHOF can delay parent changes when the new route does not provide a sufficiently large improvement.

A Sinkhole attack can also be combined with a Blackhole attack, where the Sinkhole redirects traffic towards the compromised node and the Blackhole subsequently discards it.

## 2.3 Intrusion Detection Systems (IDS) in IoT

IDS provide an additional layer of defense by monitoring network activity and identifying behavior that deviates from expected protocol operation.

IDS approaches can be deployed directly on network nodes or at a centralized monitoring point. They can also rely on different detection strategies, such as predefined protocol rules or the identification of anomalous behavior.

In this work, the IDS is based on RPL routing information and focuses on identifying inconsistencies in the observed routing hierarchy. In particular, Rank information advertised through DIO messages and parent relationships reported through DAO messages can be used to assess whether the resulting topology is consistent with the expected RPL hierarchy.

# 3 Objectives

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# 4 System setup

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# 5 Experiments

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# 6 Results

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# Acronyms

**6LoWPAN** IPv6 over Low-Power Wireless Personal Area Networks. 2

**DAO** Destination Advertisement Object. 3

**DAO-ACK** Destination Advertisement Object Acknowledgment. 3

**DIO** DODAG Information Object. 2, 3

**DIS** DODAG Information Solicitation. 2

**DODAG** Destination-Oriented Directed Acyclic Graph. 2, 3

**ETX** Expected Transmission Count. 2

**ICMPv6** Internet Control Message Protocol for IPv6. 2

**IDS** Intrusion Detection System. 2, 3

**IETF** Internet Engineering Task Force. 2

**IoT** Internet of Things. 2

**LLN** Low-Power and Lossy Network. 2

**MRHOF** Minimum Rank with Hysteresis Objective Function. 2, 3

**OF** Objective Function. 2, 3

**OF0** Objective Function Zero. 2

**RPL** Routing Protocol for Low-Power and Lossy Networks. 2, 3

# References

[1] Behnam Farzaneh, Mohammad Ali Montazeri, and Shahram Jamali. An anomaly-based ids for detecting attacks in rpl-based internet of things. pages 61–66, 2019.